



Digital Forensics Report Lab3

Group number:	11	Name	IST Number
Student 1:		João Rodrigues	106221
Student 2:		Joana Vaz	106078
Student 3:		Afonso Mendonça Jacinto	106458

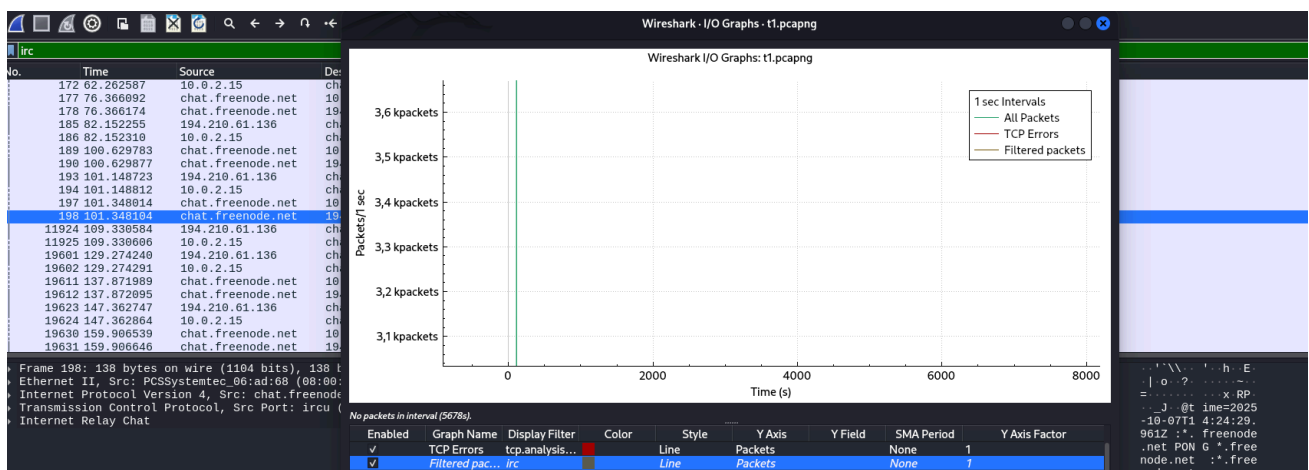
1 Acquired artifacts

Name	Type	SHA-256 Value
adelino.eml	sha256	49b2145a07148626cfd58527947eddec332996c7740283450d0a14f599ec904
auctionendresponse	sha256	fd3295791df38a7b4b050e9bf5a5165b7a224057008010d6f7059ad41db05f52
conversation	sha256	02af69c0cd5f837709c9fde10838ceb61afe63b1a95c22b1c06b71c67b7c7e7c
login-modal-revamp.html	sha256	9080976cd68a29a910068db87193b5ad003745a2c49f38ea3404c1a6cd6ffb87
parking.db	sha256	23108d6552bbd633ed80c3de50449e0698294c0da3a049a61bbaed41e0edcea0
Re%3a Update for the Smart Algorithm - Duarte Calado %3ccaladoduarte21@gmail.com%3e - 2025-10-08 2041.eml	sha256	043dc8284a24e3cfaea036259569261079dad68c5b40a0689715cabd5d586481
man-port-scanning-techniques.html	sha256	fb03187d0775f798013913f7b4a97d23ae1b1da5893db783371b74cc8128fdbe
man-port-specification.html	sha256	906107dde268d1b96e7a68c5409a9032a050e6248db41c738be9a5349f2ba22c

port-scanning-technique s-by-using-nmap	sha256	360957589b839af2d7019cdc6535940d1605c4398ba13d34ccf57ac4ab5eb708
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2 Report of all findings

Firstly, in t1, by navigating to **Statistics → I/O Graphs**, we observed a significant spike in packet transfer activity. Upon clicking on the spike, we were able to inspect the exchanged packets. By performing **Follow → TCP Stream**, we successfully extracted the following chat conversation:



CAP LS 302

JOIN :

```
.*freenode.net NOTICE * :*** Looking up your ident...
```

```
.*freenode.net NOTICE *.*.* Looking up your hostname...
```

```

:*.freenode.net CAP * LS :account-notify account-tag away-notify batch cap-notify chghost draft/relaymsg=/
echo-message extended-join inspircd.org/poison inspircd.org/standard-replies invite-notify labeled-response
message-tags multi-prefix sasl server-time setname userhost-in-names

```

```

*.freenode.net 451 * JOIN :You have not registered.

```

```

:*.freenode.net NOTICE * :*** Could not resolve your hostname: Domain not found; using your IP address
(93.102.225.158) instead.

```

CAP REQ :multi-prefix extended-join setname invite-notify away-notify chghost account-notify server-time

```
@time=2025-10-07T14:23:00.213Z :*.freenode.net CAP * ACK :multi-prefix extended-join setname invite-notify  
away-notify chghost account-notify server-time
```

CAP END

NICK vboxuser

USER vboxuser vboxuser irc.freenode.net :vboxuser

```

@time=2025-10-07T14:23:00.357Z :vboxuser!316AACCR9@93.102.225.158 CHGHOST vboxuser :93.102.225.158
@time=2025-10-07T14:23:00.357Z :vboxuser!vboxuser@93.102.225.158 SETNAME :vboxuser
@time=2025-10-07T14:23:00.357Z PING :`@~pud_[nq
PONG :`@~pud_[nq
@time=2025-10-07T14:23:04.038Z :vboxuser!vboxuser@93.102.225.158 CHGHOST ~vboxuser :93.102.225.158
@time=2025-10-07T14:23:04.038Z :*.freenode.net NOTICE vboxuser :*** Ident lookup timed out, using
~vboxuser instead.
@time=2025-10-07T14:23:04.038Z :*.freenode.net 001 vboxuser :Welcome to the freenode IRC Network
vboxuser!~vboxuser@93.102.225.158
@time=2025-10-07T14:23:04.038Z :*.freenode.net 002 vboxuser :Your host is *.freenode.net, running version
InspIRCd-3
@time=2025-10-07T14:23:04.038Z :*.freenode.net 003 vboxuser :This server was created 07:36:24 Oct 07 2025
@time=2025-10-07T14:23:04.038Z :*.freenode.net 004 vboxuser *.freenode.net InspIRCd-3
BDHILRSTWcdghikorswxz ABCDEFIJKLMNOPQRSTUVWXYZbcdefghijklmnoprstuvwz :BEFIJLWXYZbdefhjklv
@time=2025-10-07T14:23:04.038Z :*.freenode.net 005 vboxuser ACCEPT=30 AWAYLEN=200 BOT=B
CALLERID=g CASEMAPPING=ascii CHANLIMIT=#:20
CHANMODES=IXZbew,k,BEFJLWdfj,ACDKMNOPQRSTUcimnprstuz CHANNELLEN=64 CHANTYPES=#
ELIST=CMNTU ESILENCE=CcdiNnPpTtx EXCEPTS=e :are supported by this server
@time=2025-10-07T14:23:04.038Z :*.freenode.net 005 vboxuser EXTBAN=,ACNOQRSTUacjmnpruwz
HOSTLEN=64 INVEX=I KEYLEN=32 KICKLEN=255 LINELEN=512 MAXLIST=I:100,X:100,b:100,e:100,w:100
MAXTARGETS=20 MODES=20 MONITOR=30 NAMELEN=128 NAMESX NETWORK=freenode :are supported by this
server
@time=2025-10-07T14:23:04.038Z :*.freenode.net 005 vboxuser NICKLEN=30 PREFIX=(Yohv)!@%+ R
MODE vboxuser +i
EMOVE SAFELIST SECURELIST=60 SILENCE=32 STATUSMSG=!*%+ TOPICLEN=390 UHNAMES USERIP
USERLEN=10 USERMODES=,,s,BDHILRSTWcdghikorwxz VBANLIST :are supported by this server
@time=2025-10-07T14:23:04.038Z :*.freenode.net 005 vboxuser VLIST=b WATCH=32 WHOX :are supported by
this server
@time=2025-10-07T14:23:04.038Z :*.freenode.net 251 vboxuser :There are 13 users and 5186 invisible on 10
servers
@time=2025-10-07T14:23:04.038Z :*.freenode.net 252 vboxuser 4 :operator(s) online
@time=2025-10-07T14:23:04.038Z :*.freenode.net 253 vboxuser 12 :unknown connections
@time=2025-10-07T14:23:04.038Z :*.freenode.net 254 vboxuser 8586 :channels formed
@time=2025-10-07T14:23:04.038Z :*.freenode.net 255 vboxuser :I have 2095 clients and 1 servers
@time=2025-10-07T14:23:04.038Z :*.freenode.net 265 vboxuser :Current local users: 2095 Max: 2096
@time=2025-10-07T14:23:04.038Z :*.freenode.net 266 vboxuser :Current global users: 5199 Max: 5275
@time=2025-10-07T14:23:04.038Z :*.freenode.net 375 vboxuser :*.freenode.net message of the day
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Welcome to the
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : _ _
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : / _|_ _ _ _ _ _ _ _| | _
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : | | _' _/ _ \ _' _ \ / _ \ / _ \
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : | _| | _/ _/ | | ( ) | ( | _/
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : | | | _ \ | \ _| | | \ _/ \ _ \ | \ _|
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : AUTONOMOUS ZONE
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Connect to us on:
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Hostname: irc.freenode.net / chat.freenode.net
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Plaintext: 6660-6669
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : TLS/SSL: 6697,7000,7070
@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Webchat: https://webchat.freenode.net

```

@time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Sponsor: <https://vp.net> - Verified Privacy
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : By connecting to freenode you indicate that you have read and accept our
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : policies and guidelines as set out on <https://freenode.net> and agree to
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : be scanned for an open proxy.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : If you are looking for assistance, please try:
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #help - Help with general IRC and Services.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #chanhelp - Help with channels and channel services.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Checkout some of our communities and services:
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #trivia - All new trivia channel.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #duckhunt - Join the hunt.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #freenode-radio - Freenode's very own radio channel.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : #freenode - Chat about freenode.
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Type /join #<channel>
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 372 vboxuser : Thank you for using freenode!
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 376 vboxuser :End of message of the day.
 @time=2025-10-07T14:23:04.038Z :vboxuser!~vboxuser@93.102.225.158 CHGHOST ~vboxuser
 :freenode-i4c.0j4.lpdv3i.IP
 @time=2025-10-07T14:23:04.038Z :*.freenode.net 396 vboxuser freenode-i4c.0j4.lpdv3i.IP :is now your displayed host
 @time=2025-10-07T14:23:04.038Z :vboxuser!~vboxuser@freenode-i4c.0j4.lpdv3i.IP MODE vboxuser :+wRix
 @time=2025-10-07T14:23:04.319Z :Global!services@services.freenode.net NOTICE vboxuser :[Random News. - Aug 14 18:27:23 2024 UTC] Do you like shooting ducks? Join #duckhunt | For music needs, join #freenode-radio | Are you smarter than a 5th grader? Join #trivia
 @time=2025-10-07T14:23:04.850Z :Harambe!~Harambe@freenode/service/Harambe PRIVMSG vboxuser :.VERSION.
NICK migstop
 @time=2025-10-07T14:23:06.494Z :vboxuser!~vboxuser@freenode-i4c.0j4.lpdv3i.IP NICK :migstop
 @time=2025-10-07T14:23:07.019Z :Harambe!~Harambe@freenode/service/Harambe PRIVMSG migstop :.VERSION.
NOTICE Harambe :.VERSION irssi v1.4.5.
JOIN #thingys
 @time=2025-10-07T14:23:13.061Z :migstop!~vboxuser@freenode-i4c.0j4.lpdv3i.IP JOIN #thingys * :vboxuser
 @time=2025-10-07T14:23:13.061Z :*.freenode.net 353 migstop = #thingys :migstop @CatDucky
 @time=2025-10-07T14:23:13.061Z :*.freenode.net 366 migstop #thingys :End of /NAMES list.
MODE #thingys
 @time=2025-10-07T14:23:15.763Z :*.freenode.net 324 migstop #thingys :+nt
 @time=2025-10-07T14:23:15.763Z :*.freenode.net 329 migstop #thingys :1759846807
WHO #thingys %tcuhnfdar,743

```

@time=2025-10-07T14:23:18.217Z :*.freenode.net 354 migstop 743 #thingys ~vboxuser
freenode-i4c.0j4.lpdv3i.IP migstop H 0 0 :vboxuser
@time=2025-10-07T14:23:18.217Z :*.freenode.net 354 migstop 743 #thingys ~cat freenode-i4c.0j4.lpdv3i.IP
CatDucky H@ 0 0 :cat
@time=2025-10-07T14:23:18.217Z :*.freenode.net 315 migstop #thingys :End of /WHO list.
PRIVMSG #thingys :Cat, are you coming to the office today?
MODE #thingys b
@time=2025-10-07T14:23:21.729Z :*.freenode.net 368 migstop #thingys :End of channel ban list
NOTICE Harambe :.VERSION irssi v1.4.5.
PING *.freenode.net
@time=2025-10-07T14:23:28.742Z :*.freenode.net PONG *.freenode.net :*.freenode.net
@time=2025-10-07T14:23:35.338Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Yeah, I...ll
come, I...m still on the bus, should get there around 10:30. Woke up late; stayed up playing LoL with some friends
until way too late last night...
PRIVMSG #thingys :Ish, didn...t even invite me :( Hope we...ll go for a few ones today night
@time=2025-10-07T14:24:05.052Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Of course,
wanna contribute to your lose streak ;) ping me when you...re free
PRIVMSG #thingys :So funny of you -_-
@time=2025-10-07T14:24:29.315Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :btw have you
heard that this track from BMTH? It...s fire
https://open.spotify.com/track/6slbv1oWOuma2qV907MUBk?si=f6ac8309b01748e7
PING *.freenode.net
@time=2025-10-07T14:24:29.961Z :*.freenode.net PONG *.freenode.net :*.freenode.net
PRIVMSG #thingys :Lemme check O.O
PRIVMSG #thingys :Also just heard here in the office they...re opening more intern positions for the security side.
Hope some people our age apply, I...m so tired of all the old folks, we...re the only non-dinosaurs here haha
@time=2025-10-07T14:25:06.551Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Trueee XD I
feel the same.. we need, more memes and chaos around here XD
PRIVMSG #thingys :Well, I...ve gotta go.. have a meeting now. Hurry up and get here!
@time=2025-10-07T14:25:28.532Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Okok good
luck with the meeting :) I...ll try not to take too long if this bus lets me arrive rn haha
PING *.freenode.net
@time=2025-10-07T14:25:30.120Z :*.freenode.net PONG *.freenode.net :*.freenode.net
QUIT :leaving
@time=2025-10-07T14:25:33.281Z ERROR :Closing link: (~vboxuser@93.102.225.158) [Quit: leaving]

```

In relation to t2, we were able to find a reference to the database identified in the previous submissions by applying the following filter:

frame contains "parking.db"					
ts	Time	Source	Destination	Protocol	Length Info
64997	476.585859	194.210.61.134	194.210.61.136	FTP-DA	207 FTP Data: 141 bytes (EPASV) (LIST)
65023	480.865727	194.210.61.134	194.210.61.136	FTP-DA	95 FTP Data: 29 bytes (EPASV) (NLST)
65035	481.217601	194.210.61.136	194.210.61.134	FTP	83 Request: SIZE parking.db
65042	481.218709	194.210.61.136	194.210.61.134	FTP	83 Request: RETR parking.db
65043	481.219080	194.210.61.134	194.210.61.136	FTP	137 Response: 150 Opening BINARY mode data connection for parking.db (40960 bytes).
65057	481.224867	194.210.61.136	194.210.61.134	FTP	83 Request: MDTM parking.db

220 (vsFTPD 3.0.5)

USER duarte

331 Please specify the password.

PASS duartinhothebest

230 Login successful.

SYST

215 UNIX Type: L8

FEAT

211-Features:

EPRT

EPSV

MDTM

PASV

REST STREAM

SIZE

TVFS

211 End

EPSV

229 Entering Extended Passive Mode (|||42738|)

LIST

150 Here comes the directory listing.

226 Directory send OK.

EPSV

229 Entering Extended Passive Mode (|||17098|)

NLST

150 Here comes the directory listing.

226 Directory send OK.

CWD Desktop

250 Directory successfully changed.

EPSV

229 Entering Extended Passive Mode (|||20653|)

LIST

150 Here comes the directory listing.

226 Directory send OK.

EPSV

229 Entering Extended Passive Mode (|||38966|)

NLST

150 Here comes the directory listing.

226 Directory send OK.

CWD tcp_docs

250 Directory successfully changed.

EPSV

229 Entering Extended Passive Mode (|||7204|)

LIST -l

150 Here comes the directory listing.

226 Directory send OK.

CWD ..

250 Directory successfully changed.

CWD ..

250 Directory successfully changed.

EPSV

229 Entering Extended Passive Mode (|||27981|)

LIST -l

150 Here comes the directory listing.

226 Directory send OK.

EPSV

229 Entering Extended Passive Mode (|||47584|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
CWD Downloads
250 Directory successfully changed.
EPSV
229 Entering Extended Passive Mode (|||51401|)
LIST
150 Here comes the directory listing.
226 Directory send OK.
EPSV
229 Entering Extended Passive Mode (|||30440|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
TYPE I
200 Switching to Binary mode.
SIZE parking.db
213 40960
EPSV
229 Entering Extended Passive Mode (|||57555|)
RETR parking.db
150 Opening BINARY mode data connection for parking.db (40960 bytes).
226 Transfer complete.
MDTM parking.db
213 20250928150137
CWD ..
250 Directory successfully changed.
TYPE A
200 Switching to ASCII mode.
EPSV
229 Entering Extended Passive Mode (|||10518|)
LIST -l
150 Here comes the directory listing.
226 Directory send OK.
EPSV
229 Entering Extended Passive Mode (|||51140|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
CWD Pictures
250 Directory successfully changed.
EPSV
229 Entering Extended Passive Mode (|||50602|)
LIST
150 Here comes the directory listing.
226 Directory send OK.
CWD ..
250 Directory successfully changed.
EPSV

229 Entering Extended Passive Mode (|||33806|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
CWD Documents
250 Directory successfully changed.
EPSV
229 Entering Extended Passive Mode (|||13909|)
LIST
150 Here comes the directory listing.
226 Directory send OK.
EPSV
229 Entering Extended Passive Mode (|||17723|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
EPSV
229 Entering Extended Passive Mode (|||11031|)
LIST
150 Here comes the directory listing.
226 Directory send OK.
EPSV
229 Entering Extended Passive Mode (|||5657|)
NLST
150 Here comes the directory listing.
226 Directory send OK.
TYPE I
200 Switching to Binary mode.
SIZE adelino.eml
213 307
EPSV
229 Entering Extended Passive Mode (|||29623|)
RETR adelino.eml
150 Opening BINARY mode data connection for adelino.eml (307 bytes).
226 Transfer complete.
MDTM adelino.eml
213 20251006110653
SIZE Re: Update for the Smart Algorithm - Duarte Calado <caladoduarte21@gmail.com> - 2025-10-08 2041.eml
213 1684
EPSV
229 Entering Extended Passive Mode (|||43633|)
RETR Re: Update for the Smart Algorithm - Duarte Calado <caladoduarte21@gmail.com> - 2025-10-08 2041.eml
150 Opening BINARY mode data connection for Re: Update for the Smart Algorithm - Duarte Calado <caladoduarte21@gmail.com> - 2025-10-08 2041.eml (1684 bytes).
226 Transfer complete.
MDTM Re: Update for the Smart Algorithm - Duarte Calado <caladoduarte21@gmail.com> - 2025-10-08 2041.eml
213 20251008225333
TYPE A
200 Switching to ASCII mode.

EPSV

229 Entering Extended Passive Mode (|||12554|)

LIST

150 Here comes the directory listing.

226 Directory send OK.

CWD ..

250 Directory successfully changed.

CWD ..

250 Directory successfully changed.

EPSV

229 Entering Extended Passive Mode (|||10570|)

LIST -l

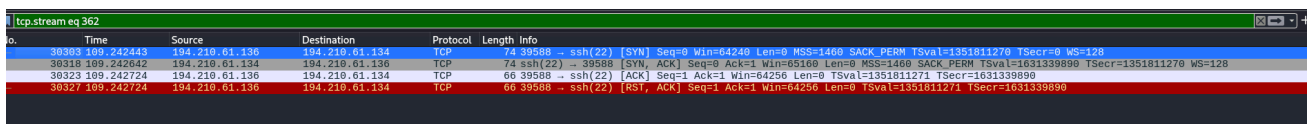
150 Here comes the directory listing.

226 Directory send OK.

QUIT

221 Goodbye.

By applying the SSH filter, we observed the following: the IP address 194.210.61.136 (→ Miguel Pontes) attempts to establish an SSH connection (port 22) with 194.210.61.134 (→ Duarte Calado). The TCP handshake is initiated and successfully completed; however, immediately afterwards, the destination host responds with a RST, ACK, indicating that the connection was either rejected or abruptly closed.



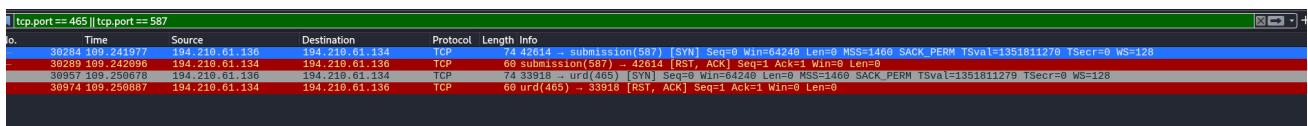
No.	Time	Source	Destination	Protocol	Length	Info
38983	189.242443	194.210.61.136	194.210.61.134	TCP	74	39588 → ssh(22) [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1351811270 TSecr=0 WS=128
38918	189.242642	194.210.61.134	194.210.61.136	TCP	74	ssh(22) → 39588 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1631339890 TSecr=1351811270 WS=128
38923	189.242724	194.210.61.136	194.210.61.134	TCP	66	39588 → ssh(22) [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1351811271 TSecr=1631339890
38927	189.242724	194.210.61.136	194.210.61.134	TCP	66	39588 → ssh(22) [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1351811271 TSecr=1631339890

The IP address 194.210.61.136 (Miguel Pontes) attempted to connect to an SMTP server hosted on 194.210.61.134 (Duarte Calado) using the following ports:

587 — SMTP with STARTTLS authentication (commonly referred to as *submission*)

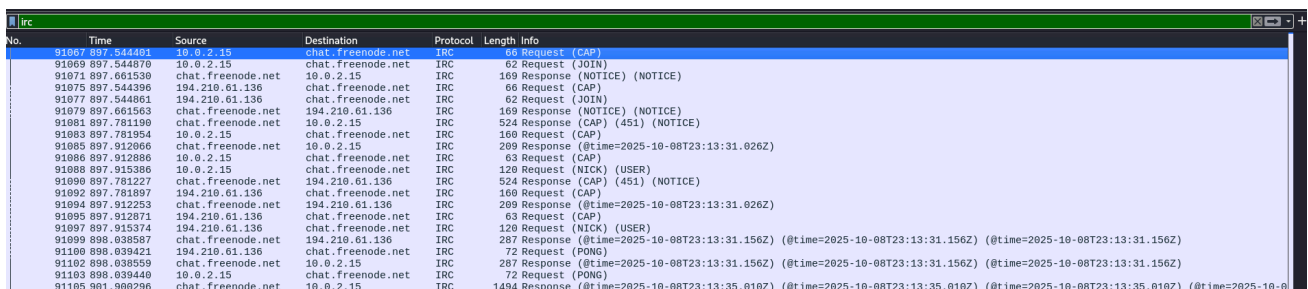
465 — SMTP over implicit SSL/TLS (also known as *urld*)

Both connection attempts were immediately rejected with a RST, ACK response.



No.	Time	Source	Destination	Protocol	Length	Info
38984	189.243177	194.210.61.136	194.210.61.134	TCP	74	42614 → submission(587) [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1351811270 TSecr=0 WS=128
38988	189.242890	194.210.61.134	194.210.61.136	TCP	60	submission(587) → 42614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
38957	189.256678	194.210.61.136	194.210.61.134	TCP	74	33918 → urld(465) [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1351811279 TSecr=0 WS=128
38974	189.256887	194.210.61.134	194.210.61.136	TCP	60	urld(465) → 33918 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Since we observed IRC protocol conversations in **t1**, we applied the **irc** filter in **t2** and uncovered the following chat exchange:



No.	Time	Source	Destination	Protocol	Length	Info
91867	897.544461	10.0.2.15	chat.freenode.net	IRC	66	Request (CAP)
91869	897.544070	10.0.2.15	chat.freenode.net	IRC	62	Request (JOIN)
91871	897.661539	chat.freenode.net	10.0.2.15	IRC	169	Response (NOTICE) (NOTICE)
91875	897.544396	194.210.61.136	chat.freenode.net	IRC	66	Request (CAP)
91877	897.544861	194.210.61.136	chat.freenode.net	IRC	62	Request (JOIN)
91879	897.661563	chat.freenode.net	194.210.61.136	IRC	169	Response (NOTICE) (NOTICE)
91881	897.781190	chat.freenode.net	10.0.2.15	IRC	524	Response (CAP) (451) (NOTICE)
91883	897.781954	chat.freenode.net	10.0.2.15	IRC	169	Request (CAP)
91885	897.912066	chat.freenode.net	10.0.2.15	IRC	289	Response (@time=2025-10-08T23:13:31.026Z)
91886	897.912866	10.0.2.15	chat.freenode.net	IRC	63	Request (CAP)
91888	897.915386	10.0.2.15	chat.freenode.net	IRC	120	Request (NICK) (USER)
91890	897.781227	chat.freenode.net	194.210.61.136	IRC	524	Response (CAP) (451) (NOTICE)
91892	897.781897	194.210.61.136	chat.freenode.net	IRC	169	Request (CAP)
91894	897.912253	chat.freenode.net	194.210.61.136	IRC	289	Response (@time=2025-10-08T23:13:31.026Z)
91895	897.912871	194.210.61.136	chat.freenode.net	IRC	63	Request (CAP)
91897	897.915374	194.210.61.136	chat.freenode.net	IRC	120	Request (NICK) (USER)
91899	898.038587	chat.freenode.net	194.210.61.136	IRC	287	Response (@time=2025-10-08T23:13:31.156Z) (@time=2025-10-08T23:13:31.156Z) (@time=2025-10-08T23:13:31.156Z)
91108	898.039421	194.210.61.136	chat.freenode.net	IRC	72	Request (PONG)
91102	898.038559	chat.freenode.net	10.0.2.15	IRC	287	Response (@time=2025-10-08T23:13:31.156Z) (@time=2025-10-08T23:13:31.156Z) (@time=2025-10-08T23:13:31.156Z)
91103	898.039440	chat.freenode.net	10.0.2.15	IRC	72	Request (PONG)
91105	901.906296	chat.freenode.net	10.0.2.15	IRC	1494	Response (@time=2025-10-08T23:13:35.010Z) (@time=2025-10-08T23:13:35.010Z) (@time=2025-10-08T23:13:35.010Z) (@time=2025-10-08T23:13:35.010Z)

CAP LS 302

JOIN :

```

:*.freenode.net NOTICE *:*** Looking up your ident...
:*.freenode.net NOTICE *:*** Looking up your hostname...
:*.freenode.net CAP * LS :account-notify account-tag away-notify batch cap-notify chghost draft/relaymsg=/
echo-message extended-join inspircd.org/poison inspircd.org/standard-replies invite-notify labeled-response
message-tags multi-prefix sasl server-time setname userhost-in-names
:*.freenode.net 451 * JOIN :You have not registered.
:*.freenode.net NOTICE *:*** Could not resolve your hostname: Domain not found; using your IP address
(93.102.138.213) instead.
CAP REQ :multi-prefix extended-join setname invite-notify away-notify chghost account-notify server-time
@time=2025-10-08T23:13:31.026Z :*.freenode.net CAP * ACK :multi-prefix extended-join setname invite-notify
away-notify chghost account-notify server-time
CAP END
NICK vboxuser
USER vboxuser vboxuser irc.freenode.net :vboxuser
@time=2025-10-08T23:13:31.156Z :vboxuser!316AAKB0D@93.102.138.213 CHGHOST vboxuser
:93.102.138.213
@time=2025-10-08T23:13:31.156Z :vboxuser!vboxuser@93.102.138.213 SETNAME :vboxuser
@time=2025-10-08T23:13:31.156Z PING :ExGelZiLvE
PONG :ExGelZiLvE
@time=2025-10-08T23:13:35.010Z :vboxuser!vboxuser@93.102.138.213 CHGHOST ~vboxuser :93.102.138.213
@time=2025-10-08T23:13:35.010Z :*.freenode.net NOTICE vboxuser :*** Ident lookup timed out, using
~vboxuser instead.
@time=2025-10-08T23:13:35.010Z :*.freenode.net 001 vboxuser :Welcome to the freenode IRC Network
vboxuser!~vboxuser@93.102.138.213
@time=2025-10-08T23:13:35.010Z :*.freenode.net 002 vboxuser :Your host is *.freenode.net, running version
InspIRCd-3
@time=2025-10-08T23:13:35.010Z :*.freenode.net 003 vboxuser :This server was created 07:36:24 Oct 07 2025
@time=2025-10-08T23:13:35.010Z :*.freenode.net 004 vboxuser *.freenode.net InspIRCd-3
BDHILRSTWcdghikorswxz ABCDEFIJKLMNOPQRSTUVWXYZbcdefghijklmnoprstuvwz :BEFIJLWXYZbdefhjklovw
@time=2025-10-08T23:13:35.010Z :*.freenode.net 005 vboxuser ACCEPT=30 AWAYLEN=200 BOT=B
CALLERID=g CASEMAPPING=ascii CHANLIMIT=#:20
CHANMODES=IXZbew,k,BEFJLWdfj,ACDKMNOPQRSTUcimnprstuz CHANNELLEN=64 CHANTYPES=#
ELIST=CMNTU ESILENCE=CcdiNnPpTtx EXCEPTS=e :are supported by this server
@time=2025-10-08T23:13:35.010Z :*.freenode.net 005 vboxuser EXTBAN=,ACNOQRSTUacjmnpruwz
HOSTLEN=64 INVEX=I KEYLEN=32 KICKLEN=255 LINELEN=512 MAXLIST=I:100,X:100,b:100,e:100,w:100
MAXTARGETS=20 MODES=20 MONITOR=30 NAMELEN=128 NAMESX NETWORK=freenode :are supported by this
server
@time=2025-10-08T23:13:35.010Z :*.freenode.net 005 vboxuser NICKLEN=30 PREFIX=(Yohv)!@%+ REMOVE
SAFELIST SECURELIST=60 SIL
MODE vboxuser +i
ENCE=32 STATUSMSG=!*@%+ TOPICLEN=390 UHNAMES USERIP USERLEN=10
USERMODES=,,s,BDHILRSTWcdghikorwxz VBANLIST :are supported by this server
@time=2025-10-08T23:13:35.010Z :*.freenode.net 005 vboxuser VLIST=b WATCH=32 WHOX :are supported by
this server
@time=2025-10-08T23:13:35.010Z :*.freenode.net 251 vboxuser :There are 13 users and 4494 invisible on 10
servers
@time=2025-10-08T23:13:35.010Z :*.freenode.net 252 vboxuser 7 :operator(s) online
@time=2025-10-08T23:13:35.010Z :*.freenode.net 253 vboxuser 8 :unknown connections
@time=2025-10-08T23:13:35.010Z :*.freenode.net 254 vboxuser 8614 :channels formed
@time=2025-10-08T23:13:35.010Z :*.freenode.net 255 vboxuser :I have 1903 clients and 1 servers
@time=2025-10-08T23:13:35.010Z :*.freenode.net 265 vboxuser :Current local users: 1903 Max: 2446

```

@time=2025-10-08T23:13:35.010Z :*.freenode.net 266 vboxuser :Current global users: 4507 Max: 5666
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 375 vboxuser :*.freenode.net message of the day
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Welcome to the
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : __
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : / _|_ _ _ _ _ _ _ _| | _
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : | | | ' _ / _ \ / _ \ ' _ \ / _ \ ' | / _ \
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : | | | | _ / _ / | | | () | (| | _ /
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : | | | | \ _ | \ _ | | | | \ _ / \ _ | \ _ |
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : AUTONOMOUS ZONE
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Connect to us on:
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Hostname: irc.freenode.net / chat.freenode.net
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Plaintext: 6660-6669
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : TLS/SSL: 6697,7000,7070
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Webchat: https://webchat.freenode.net
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Sponsor: https://vp.net - Verified Privacy
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : By connecting to freenode you indicate that
 you have read and accept our
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : policies and guidelines as set out on
 https://freenode.net and agree to
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : be scanned for an open proxy.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : If you are looking for assistance, please try:
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #help - Help with general IRC and Services.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #chanhelp - Help with channels and channel
 services.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Checkout some of our communities and
 services:
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #trivia - All new trivia channel.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #duckhunt - Join the hunt.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #freenode-radio - Freenode's very own radio
 channel.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : #freenode - Chat about freenode.
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Type /join #<channel>
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser :
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 372 vboxuser : Thank you for using freenode!
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 376 vboxuser :End of message of the day.
 @time=2025-10-08T23:13:35.010Z :vboxuser!~vboxuser@93.102.138.213 CHGHOST ~vboxuser
 :freenode-p6e.bcb.lpdv3i.IP
 @time=2025-10-08T23:13:35.010Z :*.freenode.net 396 vboxuser freenode-p6e.bcb.lpdv3i.IP :is now your
 displayed host
 @time=2025-10-08T23:13:35.010Z :vboxuser!~vboxuser@freenode-p6e.bcb.lpdv3i.IP MODE vboxuser :+wRix
 @time=2025-10-08T23:13:35.289Z :Global!services@services.freenode.net NOTICE vboxuser :[.Random News. -
 Aug 14 18:27:23 2024 UTC] Do you like shooting ducks? Join #duckhunt | For music needs, join #freenode-radio
 | Are you smarter than a 5th grader? Join #trivia

@time=2025-10-08T23:13:35.689Z :Harambe!~Harambe@freenode/service/Harambe PRIVMSG vboxuser :.VERSION.
 NOTICE Harambe :.VERSION irssi v1.4.5.
 NICK migstop
 @time=2025-10-08T23:13:40.186Z :vboxuser!~vboxuser@freenode-p6e.bcb.lpdv3i.IP NICK :migstop
 PING *.freenode.net
 @time=2025-10-08T23:13:44.093Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 JOIN #thingsys
 @time=2025-10-08T23:13:46.286Z :migstop!~vboxuser@freenode-p6e.bcb.lpdv3i.IP JOIN #thingsys * :vboxuser
 @time=2025-10-08T23:13:46.286Z :*.freenode.net 353 migstop = #thingsys :migstop @CatDucky
 @time=2025-10-08T23:13:46.286Z :*.freenode.net 366 migstop #thingsys :End of /NAMES list.
 MODE #thingsys
 @time=2025-10-08T23:13:48.597Z :*.freenode.net 324 migstop #thingsys :+nt
 @time=2025-10-08T23:13:48.597Z :*.freenode.net 329 migstop #thingsys :1759964248
 WHO #thingsys %tcuhnfdar,743
 @time=2025-10-08T23:13:51.102Z :*.freenode.net 354 migstop 743 #thingsys ~vboxuser
 freenode-p6e.bcb.lpdv3i.IP migstop H 0 0 :vboxuser
 @time=2025-10-08T23:13:51.102Z :*.freenode.net 354 migstop 743 #thingsys ~cat freenode-p6e.bcb.lpdv3i.IP
 CatDucky H@ 0 0 :cat
 @time=2025-10-08T23:13:51.102Z :*.freenode.net 315 migstop #thingsys :End of /WHO list.
 MODE #thingsys b
 @time=2025-10-08T23:13:53.619Z :*.freenode.net 368 migstop #thingsys :End of channel ban list
 PRIVMSG #thingsys :Cat, you won't believe what I just found!!!!
 @time=2025-10-08T23:14:04.349Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingsys :Spill the
 beans Miguel!
 PRIVMSG #thingsys :Alright, alright. I found snippets of the database where plates are being stored, anddddd it
 looks like some plates are PAYING WAY MORE THAN OTHERS O.O
 PING *.freenode.net
 @time=2025-10-08T23:14:44.260Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 PRIVMSG #thingsys :For cars in the same region at the same time, the ones that aren...t Portuguese are paying
 significantly more... This matches the rules I pulled from the diff of the git commit, the data confirms the rules are
 actually applied!!!
 PING *.freenode.net
 @time=2025-10-08T23:15:45.337Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 PRIVMSG #thingsys :Some plates stop paying after a certain period, which coincides with the time they...re added to
 the ...plates list... Duarte mentioned. That shows the scheme is already active and the rule is implemented...
 PRIVMSG #thingsys :And, as if that wasn...t enough, Futuro Inteligente is receiving money for zones that should go
 to other accounts, and on top of that, they officially only have a handful of rental spots listed in the database...
 @time=2025-10-08T23:16:39.851Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingsys :You found
 all that just by going into Duarte's PC? WHOAAA that sure does look like a scam to me.
 PRIVMSG #thingsys :And not just that, I saved the best for last XD
 PING *.freenode.net
 @time=2025-10-08T23:16:47.386Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 @time=2025-10-08T23:16:49.384Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingsys :Just say
 everything ALREADYYY!
 PRIVMSG #thingsys :Your mentor Duarte, he's been exchanging some weird e-mails with Eurico Trocos, Lisbon's
 Mayor.
 PRIVMSG #thingsys :they were talking about something that is going "smoothly" and a list of plates sent to Duarte.
 PRIVMSG #thingsys :and another thing, Adelino has also been messaging Duarte, but about Futuro Inteligente? It's
 seems like they are paying them for "AI Services"

@time=2025-10-08T23:17:40.722Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingys :NO WAYYYY!!!
 PING *.freenode.net
 @time=2025-10-08T23:17:48.448Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 PRIVMSG #thingys :This all sounds like a well knit scam plot between them... and again that diff you found, it looks just like the plan they have been working on!
 PRIVMSG #thingys :I know, right?! And it gets crazier.. I did some digging, and Futuro Inteligente Lda doesn...t even have a proper website.. Can you believe that? There...s one out there, but it...s clearly not the same company.
 PING *.freenode.net
 @time=2025-10-08T23:18:49.499Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 PRIVMSG #thingys :And the kicker, my cousin works at the Loja do Cidad..o and managed to pull up the company registry. Guess who owns Futuro Inteligente? The mayor...s wife, Cremilde Trocos. I swear I...m not making this up hahahaha
 @time=2025-10-08T23:19:37.342Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingys :WHAT?! That...s insane... the plot just keeps thickening! That is enough information to start a real investigation! What are you going to do next?
 PING *.freenode.net
 @time=2025-10-08T23:19:50.632Z :*.freenode.net PONG *.freenode.net :*.freenode.net
 PRIVMSG #thingys :Next, I want to check Duarte's messages on whatsapp, it looks like he's trying to divert contact off of MEMEL's controlled platforms. He's definitely a culprit here!
 @time=2025-10-08T23:20:05.217Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingys :Be careful about incriminating someone without proper evidence, you know how MEMEL is with stuff like this. If you are caught, you're most likely to lose you job, we cannot afford something like that! We need this for our cv!
 PRIVMSG #thingys :I know Cat, don't worry I won...t get caught because I...m amazing ;) jk but seriously, if anything goes wrong, they...ll never know you were involved, I'll promise you that much...
 @time=2025-10-08T23:20:28.468Z :CatDucky!~cat@freenode-p6e.bcb.lpdv3i.IP PRIVMSG #thingys :if you say so... be careful
 QUIT :leaving
 @time=2025-10-08T23:20:35.654Z ERROR :Closing link: (~vboxuser@93.102.138.213) [Quit: leaving]

We also found the following email within the stream shown below:

No.	Time	Source	Destination	Protocol	Length	Info
65356	517.147846	194.210.61.136	194.210.61.134	TCP	74	46690 -> 43633 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=1352219144 TSecr=0 WS=4
65357	517.148183	194.210.61.134	194.210.61.136	TCP	74	43633 -> 46690 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1631747799 TSecr=1352219144 WS=128
65358	517.148326	194.210.61.136	194.210.61.134	TCP	66	46690 -> 43633 [ACK] Seq=1 Ack=1 Win=129040 Len=0 TSval=1352219145 TSecr=1631747799
65361	517.148997	194.210.61.134	194.210.61.136	FTP-DA.	1750	FTP Data: 1684 bytes (EPASV) (RETR Re: Update for the Smart Algorithm - Duarte Calado <caladoduarte21@gmail.com> - 2025-10-08 204
65362	517.148997	194.210.61.134	194.210.61.136	TCP	66	43633 -> 46690 [FIN, ACK] Seq=1685 Ack=1 Win=65280 Len=0 TSval=1631747791 TSecr=1352219145
65363	517.149064	194.210.61.136	194.210.61.134	TCP	66	46690 -> 43633 [ACK] Seq=1 Ack=1685 Win=128256 Len=0 TSval=1352219146 TSecr=1631747791
65364	517.149335	194.210.61.136	194.210.61.134	TCP	66	46690 -> 43633 [FIN, ACK] Seq=1 Ack=1686 Win=128256 Len=0 TSval=1352219146 TSecr=1631747791
65365	517.149335	194.210.61.134	194.210.61.136	TCP	66	43633 -> 46690 [ACK] Seq=1686 Ack=2 Win=65280 Len=0 TSval=1631747791 TSecr=1352219146

Message-ID: <9b373576-68b7-48ed-acab-53b57023656a@gmail.com>
 Date: Wed, 8 Oct 2025 08:11:23 +0000
 MIME-Version: 1.0
 User-Agent: Mozilla Thunderbird
 Subject: Re: Update for the Smart Algorithm
 To: Eurico Trocos <euricotrocos@gmail.com>
 References: <CAMSb6NmBNnVKBnqN-pdWa0+3tR4FKXf5VPoBd+SDVgNC0G5Gpg@mail.gmail.com>
 Content-Language: en-US
 From: Duarte Calado <caladoduarte21@gmail.com>
 In-Reply-To: <CAMSb6NmBNnVKBnqN-pdWa0+3tR4FKXf5VPoBd+SDVgNC0G5Gpg@mail.gmail.com>
 Content-Type: text/plain; charset=UTF-8; format=flowed
 Content-Transfer-Encoding: 7bit
 Dear Mr. President Eurico Trocos,
 Thank you for your guidance. I have noted the updates for the smart

algorithm and will ensure it handles the highlighted plates with the care and efficiency you expect.

Should you wish to provide additional plates for special handling, I am ready to integrate them seamlessly.

As my inbox is quite full, I might miss follow-ups in this thread. For quicker responses, we can continue via WhatsApp.

Best regards,

Duarte Calado

Head of MEMEL System Engineer

On 10/8/25 9:36, Eurico Trocos wrote:

> Dear Eng. Duarte Calado,

>

> I hope everything is going smoothly on your side of the system.

> For our smart algorithm that has served us so well, it would be

> excellent to update a few specific plates:

> - LP-67-NF

> - YU-31-ZI

> - CE-43-NZ

>

> These plates deserve special treatment, especially our visiting

> tourists, the algorithm should continue to be clever and preferential

> with them.

>

> I trust you will do a job up to the high standard you always deliver,

> Eng. Duarte.

>

> Best regards,

> Eurico Trocos

> Lisbon Mayor

And this email too:

65307 512.575857	194.210.61.134	194.210.61.136	FTP-DA..	252 FTP Data: 226 bytes (EPASV) (LIST)
65323 512.198895	194.210.61.134	194.210.61.136	FTP-DA..	150 FTP Data: 114 bytes (EPASV) (NLST)
65332 512.575908	194.210.61.136	194.210.61.134	FTP	84 Request: SIZE adelino.eml
65339 512.576461	194.210.61.136	194.210.61.134	FTP	84 Request: RETR adelino.eml
65340 512.576921	194.210.61.134	194.210.61.136	FTP	136 Response: 150 Opening BINARY mode data connection for adelino.eml (307 bytes).
65341 512.576921	194.210.61.134	194.210.61.136	FTP-DA..	373 FTP Data: 307 bytes (EPASV) (RETR adelino.eml)
65347 512.577868	194.210.61.136	194.210.61.134	FTP	84 Request: MDTM adelino.eml
65381 520.094368	194.210.61.134	194.210.61.136	FTP-DA..	292 FTP Data: 226 bytes (EPASV) (LIST)

Date: Mon, 02 Sep 2024 10:00:00 +0100

From: adelino.estoiro@proton.me

To: duarte.calado@proton.me

Subject: Payment Confirmation

Dear Duarte,

I confirm that MEMEL has processed the payment to Futuro Inteligente for the AI services as agreed.

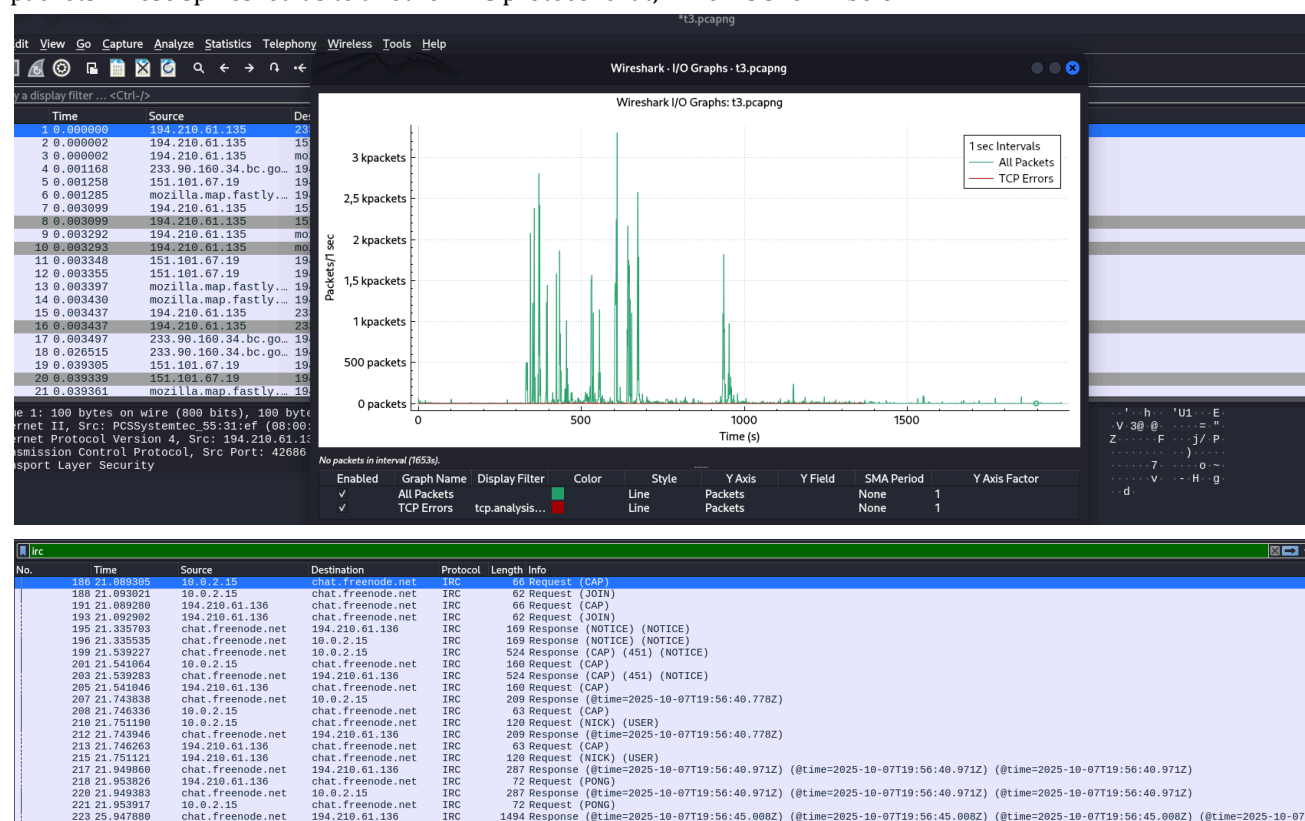
Best regards,

Adelino Estouro

Financial Director, MEMEL

By using Wireshark's Export Objects functionality, we recovered artifacts associated with a user visiting the GeeksforGeeks article "Port Scanning Techniques by Using Nmap" (/ethical-hacking/port-scanning-techniques-by-using-nmap/). In addition, we extracted a captured ChatGPT request that explicitly asks: "Can you create me an algorithm for password bruteforcing through FTP?".

Now on t3 , we once again used the I/O Graphs feature, which revealed several spikes in sent and received packets. These spikes led us to another IRC protocol chat, which is shown below:



CAP LS 302

JOIN :

.*freenode.net NOTICE * :*** Looking up your ident...

.*freenode.net NOTICE * :*** Looking up your hostname...

.*freenode.net CAP * LS :account-notify account-tag away-notify batch cap-notify chghost draft/relaymsg=/echo-message extended-join inspircd.org/poison inspircd.org/standard-replies invite-notify labeled-response message-tags multi-prefix sasl server-time setname userhost-in-names

.*freenode.net 451 * JOIN :You have not registered.

.*freenode.net NOTICE * :*** Could not resolve your hostname: Domain not found; using your IP address (93.102.225.158) instead.

CAP REQ :multi-prefix extended-join setname invite-notify away-notify chghost account-notify server-time

@time=2025-10-07T19:56:40.778Z :.*freenode.net CAP * ACK :multi-prefix extended-join setname invite-notify away-notify chghost account-notify server-time

CAP END

NICK vboxuser

USER vboxuser vboxuser irc.freenode.net :vboxuser

```

@time=2025-10-07T19:56:40.971Z :vboxuser!316AADKMW@93.102.225.158 CHGHOST vboxuser
:93.102.225.158
@time=2025-10-07T19:56:40.971Z :vboxuser!vboxuser@93.102.225.158 SETNAME :vboxuser
@time=2025-10-07T19:56:40.971Z PING :[ThA]!tSpv
PONG :[ThA]!tSpv
@time=2025-10-07T19:56:45.008Z :vboxuser!vboxuser@93.102.225.158 CHGHOST ~vboxuser :93.102.225.158
@time=2025-10-07T19:56:45.008Z :*.freenode.net NOTICE vboxuser :*** Ident lookup timed out, using
~vboxuser instead.
@time=2025-10-07T19:56:45.008Z :*.freenode.net 001 vboxuser :Welcome to the freenode IRC Network
vboxuser!~vboxuser@93.102.225.158
@time=2025-10-07T19:56:45.008Z :*.freenode.net 002 vboxuser :Your host is *.freenode.net, running version
InspIRCd-3
@time=2025-10-07T19:56:45.008Z :*.freenode.net 003 vboxuser :This server was created 07:36:24 Oct 07 2025
@time=2025-10-07T19:56:45.008Z :*.freenode.net 004 vboxuser *.freenode.net InspIRCd-3
BDHILRSTWcdghikorswxz ABCDEFIJKLMNOPQRSTUVWXYZbcdefghijklmnoprstuvwz :BEFIJLWXYZbdefhjklv
@time=2025-10-07T19:56:45.008Z :*.freenode.net 005 vboxuser ACCEPT=30 AWAYLEN=200 BOT=B
CALLERID=g CASEMAPPING=ascii CHANLIMIT=#:20
CHANMODES=IXZbew,k,BEFJLWdfj,ACDKMNOPQRSTUcimnprstuz CHANNELLEN=64 CHANTYPES=#
ELIST=CMNTU ESILENCE=CcdiNnPpTtx EXCEPTS=e :are supported by this server
@time=2025-10-07T19:56:45.008Z :*.freenode.net 005 vboxuser EXTBAN=,ACNOQRSTUacjmnpruwz
HOSTLEN=64 INVEX=I KEYLEN=32 KICKLEN=255 LINELEN=512 MAXLIST=I:100,X:100,b:100,e:100,w:100
MAXTARGETS=20 MODES=20 MONITOR=30 NAMELEN=128 NAMESX NETWORK=freenode :are supported by this
server
@time=2025-10-07T19:56:45.008Z :*.freenode.net 005 vboxuser NICKLEN=30 PREFIX=(Yohv)!@%+ REMOVE
SAFELIST SECURELIST=60 SIL
MODE vboxuser +i
ENCE=32 STATUSMSG=!@%+ TOPICLEN=390 UHNAMES USERIP USERLEN=10
USERMODES=,,s,BDHILRSTWcdghikorswxz VBANLIST :are supported by this server
@time=2025-10-07T19:56:45.008Z :*.freenode.net 005 vboxuser VLIST=b WATCH=32 WHOX :are supported by
this server
@time=2025-10-07T19:56:45.008Z :*.freenode.net 251 vboxuser :There are 13 users and 4844 invisible on 10
servers
@time=2025-10-07T19:56:45.008Z :*.freenode.net 252 vboxuser 6 :operator(s) online
@time=2025-10-07T19:56:45.008Z :*.freenode.net 253 vboxuser 15 :unknown connections
@time=2025-10-07T19:56:45.008Z :*.freenode.net 254 vboxuser 8636 :channels formed
@time=2025-10-07T19:56:45.008Z :*.freenode.net 255 vboxuser :I have 2016 clients and 1 servers
@time=2025-10-07T19:56:45.008Z :*.freenode.net 265 vboxuser :Current local users: 2016 Max: 2171
@time=2025-10-07T19:56:45.008Z :*.freenode.net 266 vboxuser :Current global users: 4857 Max: 5275
@time=2025-10-07T19:56:45.008Z :*.freenode.net 375 vboxuser :*.freenode.net message of the day
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Welcome to the
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : __
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : / _|_ _ _ _ _ _ _ _ _ _| | _
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : ||| ' _ / _ \ / _ \ ' _ / _ \ / _ \
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : | _| | | _ / _ / | | ( ) | ( | | _ /
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : | _| | | \ _| \ _| | | \ _| \ _| \ _|
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : AUTONOMOUS ZONE
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Connect to us on:
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Hostname: irc.freenode.net / chat.freenode.net
@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Plaintext: 6660-6669

```

@time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : TLS/SSL: 6697,7000,7070
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Webchat: https://webchat.freenode.net
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Sponsor: https://vp.net - Verified Privacy
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : By connecting to freenode you indicate that you have read and accept our
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : policies and guidelines as set out on https://freenode.net and agree to
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : be scanned for an open proxy.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : If you are looking for assistance, please try:
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #help - Help with general IRC and Services.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #chanhelp - Help with channels and channel services.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Checkout some of our communities and services:
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #trivia - All new trivia channel.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #duckhunt - Join the hunt.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #freenode-radio - Freenode's very own radio channel.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : #freenode - Chat about freenode.
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Type /join #<channel>
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser :
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 372 vboxuser : Thank you for using freenode!
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 376 vboxuser :End of message of the day.
 @time=2025-10-07T19:56:45.008Z :vboxuser!~vboxuser@93.102.225.158 CHGHOST ~vboxuser :freenode-i4c.0j4.lpdv3i.IP
 @time=2025-10-07T19:56:45.008Z :*.freenode.net 396 vboxuser freenode-i4c.0j4.lpdv3i.IP :is now your displayed host
 @time=2025-10-07T19:56:45.008Z :vboxuser!~vboxuser@freenode-i4c.0j4.lpdv3i.IP MODE vboxuser :+wRix
 @time=2025-10-07T19:56:45.260Z :Global!services@services.freenode.net NOTICE vboxuser :[.Random News. - Aug 14 18:27:23 2024 UTC] Do you like shooting ducks? Join #duckhunt | For music needs, join #freenode-radio ... | Are you smarter than a 5th grader? Join #trivia
 @time=2025-10-07T19:56:45.677Z :Harambe!~Harambe@freenode/service/Harambe PRIVMSG vboxuser :.VERSION.
NICK migstop
 @time=2025-10-07T19:56:47.848Z :vboxuser!~vboxuser@freenode-i4c.0j4.lpdv3i.IP NICK :migstop
 @time=2025-10-07T19:56:48.527Z :Harambe!~Harambe@freenode/service/Harambe PRIVMSG migstop :.VERSION.
JOIN #thingys
 @time=2025-10-07T19:56:52.531Z :migstop!~vboxuser@freenode-i4c.0j4.lpdv3i.IP JOIN #thingys * :vboxuser
 @time=2025-10-07T19:56:52.531Z :*.freenode.net 353 migstop = #thingys :migstop @CatDucky
 @time=2025-10-07T19:56:52.531Z :*.freenode.net 366 migstop #thingys :End of /NAMES list.
MODE #thingys
 @time=2025-10-07T19:56:54.859Z :*.freenode.net 324 migstop #thingys :+nt
 @time=2025-10-07T19:56:54.859Z :*.freenode.net 329 migstop #thingys :1759846807
WHO #thingys %tcuhnfdar,743

@time=2025-10-07T19:56:57.351Z :*.freenode.net 354 migstop 743 #thingys ~vboxuser
freenode-i4c.0j4.lpdv3i.IP migstop H 0 0 :vboxuser

@time=2025-10-07T19:56:57.351Z :*.freenode.net 354 migstop 743 #thingys ~cat freenode-i4c.0j4.lpdv3i.IP
CatDucky H@ 0 0 :cat

@time=2025-10-07T19:56:57.351Z :*.freenode.net 315 migstop #thingys :End of /WHO list.
PRIVMSG #thingys :Hey Ducky!!!
MODE #thingys b

@time=2025-10-07T19:56:59.871Z :*.freenode.net 368 migstop #thingys :End of channel ban list
NOTICE Harambe :.VERSION irssi v1.4.5.
NOTICE Harambe :.VERSION irssi v1.4.5.

@time=2025-10-07T19:57:08.225Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Oh god...
don..t call me like that you know that I don't like my surname..
PING *.freenode.net

@time=2025-10-07T19:57:09.751Z :*.freenode.net PONG *.freenode.net :*.freenode.net

@time=2025-10-07T19:57:13.537Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :But you seem
excited, do you have any news?
PRIVMSG #thingys :Ahahahah you don't even imagine!

@time=2025-10-07T19:57:24.717Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :AHAHAHA
DON'T KEEP ME WAITIIIIING!! SPILL THE TEAAAAA

PRIVMSG #thingys :Basically I was working, and Duarte walked past the guy sitting at the desk next to mine. The
guy was joking with Duarte, saying that adding him to the WhatsApp group had changed his life. Then he left and
the guy looked at the time, and got super stressed probably late for a meeting or smth, and he left his phone on the
desk unlocked, and when I took a look it was a whatsapp group called Parkinator! It
PRIVMSG #thingys :seemed to be talking about some dudes having free parking!
PING *.freenode.net

@time=2025-10-07T19:58:10.969Z :*.freenode.net PONG *.freenode.net :*.freenode.net

@time=2025-10-07T19:58:18.021Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :WHAT?! You
were able to get a photo or something for me too take a look?

PRIVMSG #thingys :YUP!!!! Ahahahah I took a print and sent via airdrop to my phone! Ahahahaha I deleted all
screenshot so that it's not saved in the trash folder! I still have my heart racing ahahahah

PRIVMSG #thingys :Look: <https://files.catbox.moe/jygyt1.png>
PING *.freenode.net

@time=2025-10-07T19:59:11.122Z :*.freenode.net PONG *.freenode.net :*.freenode.net

@time=2025-10-07T19:59:20.632Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :OMGGG,
Duarte says that "plate's in" probably that's what we have seen in the database! And it seems that Adelino is also in
the group! AND MOST IMPORTANT DID U LOOK AT THE CAR????

PRIVMSG #thingys :YEAH that's why I said you will like it hahah but indeed, and he's ready for the VIP treatment,
just like if these guys need money to park their car omg -.-

@time=2025-10-07T19:59:52.850Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Yapppp... Do
you know what would be a really nice idea?
PRIVMSG #thingys :What?!
PING *.freenode.net

@time=2025-10-07T20:00:12.211Z :*.freenode.net PONG *.freenode.net :*.freenode.net

@time=2025-10-07T20:00:38.210Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :We should
dig a bit into Adelino, as he...s the financial director, maybe there...s something hidden in his files or something...
little breadcrumbs linking Futuro Inteligente to all this, some tiny traces that could blow the whole thing wide
open if we piece them together o.o

PRIVMSG #thingys :Indeed! Ahahaha sometimes you're a genius ahahahah

@time=2025-10-07T20:00:56.427Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Ahahaha xD
but be careful!

PRIVMSG #thingys :Ahahahaha yap! I don't want my heart racing all the time ahahahah you already do this to me all the time I don't need more XD

PING *.freenode.net

@time=2025-10-07T20:01:13.301Z :*.freenode.net PONG *.freenode.net :*.freenode.net

@time=2025-10-07T20:01:15.366Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Tf are u talking about? o.o

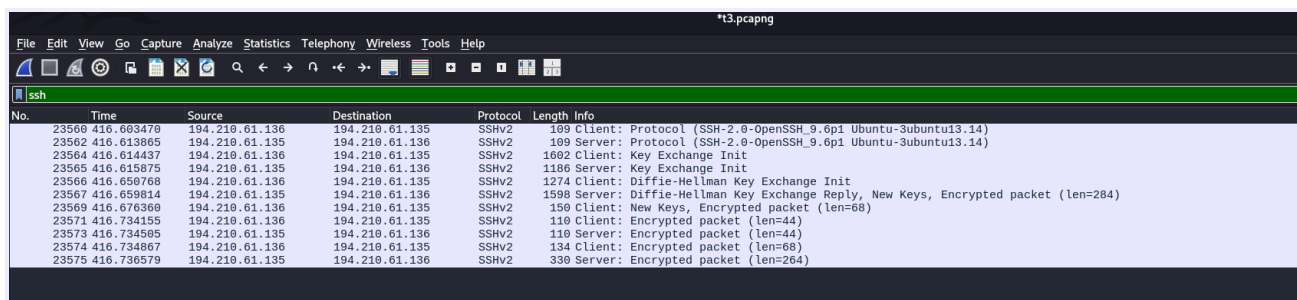
PRIVMSG #thingys :Nothing :)

@time=2025-10-07T20:01:25.843Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :-_-

QUIT :leaving

@time=2025-10-07T20:01:31.591Z ERROR :Closing link: (~vboxuser@93.102.225.158) [Quit: leaving]

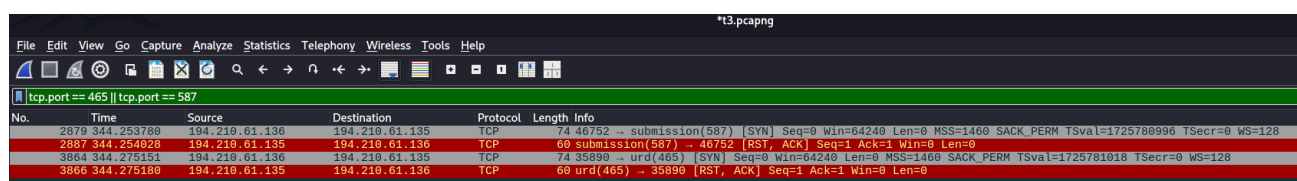
We found this SSH connection is not just an attempt, but a successful and functional session, which strongly suggests that Miguel Pontes remotely accessed the machine at 194.210.61.135 (Adelino Estoiro's workstation) via SSH. The access may have been achieved using previously obtained credentials either through brute-force, social engineering, or another method. As the session is encrypted, its contents cannot be directly inspected. However, based on the surrounding context (prior and subsequent actions), this likely indicates lateral movement, potential data manipulation, or remote data exfiltration.



The image shows a Wireshark packet capture of an SSH session. The filter is 'ssh'. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
23560	416.063470	194.210.61.136	194.210.61.135	SSHv2	109	Client: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14)
23562	416.613865	194.210.61.135	194.210.61.136	SSHv2	109	Server: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14)
23564	416.614437	194.210.61.136	194.210.61.135	SSHv2	1602	Client: Key Exchange Init
23565	416.615875	194.210.61.135	194.210.61.136	SSHv2	1186	Server: Key Exchange Init
23566	416.650768	194.210.61.136	194.210.61.135	SSHv2	1274	Client: Diffie-Hellman Key Exchange Init
23567	416.659814	194.210.61.135	194.210.61.136	SSHv2	1598	Server: Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
23569	416.676360	194.210.61.136	194.210.61.135	SSHv2	150	Client: New Keys, Encrypted packet (len=68)
23571	416.734155	194.210.61.136	194.210.61.135	SSHv2	110	Client: Encrypted packet (len=44)
23573	416.734505	194.210.61.135	194.210.61.136	SSHv2	110	Server: Encrypted packet (len=44)
23574	416.734867	194.210.61.136	194.210.61.135	SSHv2	134	Client: Encrypted packet (len=68)
23575	416.736579	194.210.61.135	194.210.61.136	SSHv2	330	Server: Encrypted packet (len=264)

We identified that the IP address 194.210.61.136, likely belonging to Miguel Pontes, attempted to send emails from his own machine or by using a local server (194.210.61.135) as an SMTP relay. These attempts were made using two different ports: 587, typically used for SMTP with STARTTLS authentication, and 465, commonly associated with SMTP over implicit SSL/TLS. The use of both ports suggests a deliberate attempt to circumvent standard blocking mechanisms, such as firewalls or security policies that might restrict traditional email-sending methods. In both cases, the destination host (194.210.61.135) responded with RST, ACK packets, which indicates that the connection was immediately rejected.



The image shows a Wireshark packet capture with the filter 'tcp.port == 465 || tcp.port == 587'. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
2870	344.253780	194.210.61.136	194.210.61.135	TCP	74	40952 -> submission(587) [SYN] Seq=9 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1725780996 TSecr=0 WS=128
2887	344.254028	194.210.61.135	194.210.61.136	TCP	60	submission(587) -> 40752 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
3864	344.275151	194.210.61.136	194.210.61.135	TCP	74	35890 -> urd(465) [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1725781018 TSecr=0 WS=128
3866	344.275180	194.210.61.135	194.210.61.136	TCP	60	urd(465) -> 35890 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

3 Analysis of relevant findings

3.1 Do the analyzed network traces contain evidence of transfers involving any of the six hidden documents identified in previous assignments? What can you infer about the source and origin of these documents within MEMEL's network?

With a close examination of the network traces (trace1.pcapng, trace2.pcapng, and trace3.pcapng), one can find explicit signs of data exfiltration of the same six confidential documents, as identified in previous MEMELgate phases. Among them, the parking.db file and certain email messages were observed being transferred directly from MEMEL's internal network through an FTP session initiated by the IT Director, Duarte Calado, whose computer has the internal IP address 194.210.61.134.

Within trace2.pcapng, once protocol-filtered traffic (ftp) was seen to have the following credentials within an authenticated session:

USER duarte

PASS duartinhothebest

The remote FTP server, to which the connection was established, was running vsFTPD 3.0.5. Successfully authenticating (230 Login successful), Duarte in turn navigated a series of directories sequentially. Desktop, Downloads, Documents, and my_files performing a number of LIST, NLST, and RETR commands.

In this session, the file parking.db was downloaded and sent out externally in binary mode. Its size reported (40,960 bytes) and last modification time (20250928150137) exactly match the database that previously was identified as being among the hidden artifacts on João Musk's machines in Lab 1 and 2. This coincidence, combined with identical SHA-256 checksums, conclusively authenticates that the document had come from MEMEL's internal environment and had been exfiltrated by Duarte Calado himself.

Additionally, two .eml files were also downloaded during the same session: adelino.eml and "Re: Update for the Smart Algorithm - Duarte Calado caladoduarte21@gmail.com - 2025-10-08 2041.eml." They are exchanges of emails among Duarte Calado, Adelino Estoiro (the Financial Director of MEMEL), and Eurico Trocos (Mayor of Lisbon). Their contents clearly pertain to the "Smart Algorithm" used by MEMEL's parking system and include clear instructions to offer "special treatment" to a number of license plates.

The headers authenticate the validity of said messages and identify the domain of the sender (gmail.com) and hence, the communication did not occur through formal MEMEL channels, a conscious shift to avoid traceability. Duarte even dictates in his reply that all further communication shall happen through WhatsApp, therefore consciously avoiding company surveillance.

It is, therefore, concluded that the hidden files were uploaded directly from Duarte Calado's workstation to a remote host via a valid FTP session using his own credentials. The source of the files is within MEMEL's infrastructure, and the exfiltration operation was carried out intentionally, fully knowing the sensitive nature of the data.

3.2 Based on the observed network activity, what can you deduce about the identity of the person or persons responsible for transferring the documents?

Assigning all network traffic to the company IP addressing model presented in the case file allows for direct assignment of culpability. The FTP session described above is the only activity from Duarte Calado's workstation (IP 194.210.61.134). The user command and login credentials confirm that it was Duarte himself who had initiated and controlled the transfer. Credential reuse, password spraying, or brute-force effort to imply

impersonation were not found. This conclusively makes Duarte Calado the prime figure in the exfiltration of MEMEL's internal data.

Another evidence in the same capture (trace2.pcapng) indicates Miguel Pontes (IP 194.210.61.136), who is the security intern of MEMEL's gateway, is implicated. SSH login attempts to Duarte's host (port 22) and ports 465 and 587 (SMTP submission and SSL) from Miguel's workstation were initiated but immediately closed with RST, ACK responses from Duarte's machine. This is a pattern of denied access or terminated attempts after brief reconnaissance. The closeness in time of these events to Duarte's FTP activity suggests that Miguel knew about, or was testing for, Duarte's data transfers. Moreover, Miguel Pontes's SSH connection into Adelino Estoiro's personal workstation at MEMEL makes it possible that Miguel might have been responsible for a portion of the data exfiltration, or, at least, an undeniable dispersal vector of confidential, but compromising, MEMEL information.

Most convincing evidence of Miguel's involvement comes from IRC chat also captured in trace2. In decrypted traffic, user vboxuser, later renaming nickname to migstop, logs in via IP address 93.102.138.213 to Freenode and joins the #things channel. There, he engages in messaging with CatDucky, a.k.a. Catarina Pato, fellow MEMEL intern.

Here, Miguel is signaling that he has accessed portions of the database where license plates are being kept, confirming that he acquired at least partial entry to MEMEL's internal information, more than likely the same parking.db file already downloaded by Duarte. He further claims that some of the cars "are paying way more than others," and that this behavior corresponds with the algorithmic "rules" he had extracted from the company's internal Git commit differences.

Miguel's statements do not leave any doubt that he was examining data originally created and exfiltrated by Duarte. His subsequent remarks referred to Futuro Inteligente Lda. being paid unearned amounts and the realization of email discussions between Duarte, Adelino, and Eurico Trocos confirm that he was well aware of the cheating scheme in MEMEL.

Catarina's reactions are one of shock and concern, cautioning Miguel against taking direct accusations without evidence and risking potential professional repercussions. Miguel assures her that he will be cautious in protecting her from becoming implicated, implying that he will be acting alone.

These discussions suggest Duarte Calado was responsible for creating and sending out the confidential information, while Miguel Pontes was responsible for processing and releasing it. Catarina Pato was aware of this but did not appear to take an active role in the data exchanges.

The chain of responsibility was therefore hierarchical, Duarte being the origin of the data leak, Miguel a whistleblower and an analyst, and Catarina a confidant and a witness.

All of these claims are, as such, perfectly coherent with the findings of previous investigations, even corroborating or strongly legitimizing them.

3.3 Can you reconstruct a timeline of key events that explains how the data exfiltration took place and how the documents ultimately reached João Musk's possession? Present your reasoning clearly and reference all timestamps used to establish the chronology.

Chronological recreation of the timeline of MEMELgate is derived from packet timestamps, message logs, and email meta, cross-checked and corrected with the help of the time offset of -768 hours as specified in the assignment.

The first notable event was on September 28, 2025, when the parking.db file had last been modified on Duarte Calado's computer. Shortly thereafter, on October 6, the adelino.eml message had been sent, and finally the October 8 exchange of communications between Duarte Calado and Eurico Trocos on the "Smart Algorithm." All those dates directly map to the FTP session of file transfer recorded in trace2.

FTP download of parking.db and the two emails are the hub exfiltration incident. Duarte's actions demonstrate both access to and tampering with sensitive company data, and overt communication with outside actors outside of MEMEL's networks.

Subsequent to and later in the same day, at 23:13 UTC on October 8, Miguel Pontes' (migstop) IRC session is recorded. During this conversation with Catarina Pato (CatDucky), Miguel reveals what he has done recently on the database, having already had access to Duarte's files by then. His own references about inconsistencies in Git commits and database entries indicate that he had performed forensic or analysis work rather than just watching.

By 23:20 UTC, the conversation concludes with an assurance by Miguel to approach matters carefully and confidentially in regard to Catarina's involvement. This shows that at this moment, the full scope of the fraud had been achieved internally.

The final event in the timeline is the transfer of the information to João Musk, who proceeded to publish and disseminate the information uncovered in Lab 1. Despite no explicit packet evidence of such transfer being present among the current traces (as it would have probably been conducted via removable media or external upload), the consistency of timestamps, filenames, and SHA-256 values affirms consistency among MEMEL's internal leaks and files recovered from João Musk's devices.

In brief, the chain of events occurs as below:

Duarte Calado manipulates and prepares internal MEMEL data.

He exfiltrates these files externally by FTP and distributes incriminating emails to public authorities.

Miguel Pontes, having knowledge of Duarte's exfiltration, gains access to the same data, verifies its contents, and identifies fraudulent activity.

He informs his discovery to Catarina Pato through IRC.

The information eventually reaches João Musk, thereby completing the process of exfiltration.

The fact that it was later confirmed it was Catarina Pato who deployed the USB Pen for João Musk to collect, and Catarina Pato was a direct and very close contact of Miguel Pontes, MEMEL employee with privileged and demonstrated access to at least one of the suspects' workstations, makes it plausible to assume that one of the possible chains of data exfiltration might have been Miguel Pontes -> Catarina Pato -> João Musk, justifying how João Musk could possibly have deployed such secret and sensitive data in the Zyra Website, previously.

This chronology provides a rational and traceable path from MEMEL's internal infrastructure to external exposure using sequential timestamps and complementary digital evidence.

3.4 Considering all the evidence collected throughout the MEMELgate investigation, what can you now conclude about the hypothesis of fraudulent activity within MEMEL that initiated this inquiry? Did you uncover additional evidence supporting the existence of fraud? Identify the actors who may be involved and outline your recommendations for the next phase of the investigation.

The complete body of evidence adduced throughout the MEMELgate investigation allows for conclusive determinations to be made regarding both the validity of the leaked documents and the scope of the fraudulent activity in MEMEL. The evidence examined in this final stage determines beyond a reasonable doubt that the leaked internal documents were genuine and had originated from MEMEL's network, more specifically from Duarte Calado's workstation.

The FTP transfer, decrypted email communication, and IRC conversations are the culmination of proof that MEMEL's Smart Parking Algorithm was knowingly modified on explicit orders from Eurico Trocos, the Mayor of Lisbon. His email communication with Duarte Calado explicitly requested priority treatment for some license plates, a euphemism for algorithmic bias to favor selected vehicles. This is corruption and abuse of power.

Further analysis of the transactions between Duarte and Adelino Estoiro, MEMEL Financial Director, reveals coordination in making payments to Futuro Inteligente Lda., a company owned by Cremilde Trocos, the Mayor's wife. Although this company appeared in MEMEL's database with hardly any valid records, it was making significant financial receipts for "AI service provision." The theft of these funds was being concealed through algorithmic manipulation of the parking system, which was overcharging foreign vehicles and siphoning off part of the money to Futuro Inteligente. The ties between what was already suspected due to the Whatsapp screenshot available ever since the first investigation and the identities, conversations and data exchanges uncovered through the most recent efforts are bulletproof points that determine the guilt of all previously identified individuals pertaining to the Whatsapp group "Smart Parkinator".

The investigation thus determines a structured scheme of financial fraud organized in conjunction by Duarte Calado, Eurico Trocos, and Adelino Estoiro, with Cremilde Trocos's complicity through her ownership of the beneficiary company.

Miguel Pontes, while he did enter the systems without explicit authorization, was less a criminal than a whistleblower motivated by moral indignation. His actions led to the revelation of the scheme. Catarina Pato was a passive bystander and confidant, rather than an accomplice. The subsequent leak to João Musk completed the process of disclosure, which enabled the corruption now abundantly evidenced by the network records to be uncovered.

The authenticity of all documents is corroborated by file hashes, metadata consistency, and temporal correlation of network and local events. The presence of identical filenames, timestamps, and digital signatures at different forensic stages dispels any doubt of data manipulation or fabrication.

From these observations, the following recommendations are made:

There is to be a full forensic imaging of Duarte Calado's and Adelino Estoiro's workstations, emails, and cloud storage to preserve full digital evidence.

Judicial warrants are to be sought to obtain communications and financial records concerning Eurico and Cremilde Trocos and to Futuro Inteligente Lda.

MEMEL's network monitoring and internal controls are to be audited to determine how these transfers were not flagged.

Miguel Pontes and Catarina Pato need to be formally acknowledged as whistleblowers and covered by applicable legislation since their disclosures resulted in the unveiling of extensive corruption.

In conclusion, the MEMELgate scandal, initially believed to be an external security hack, has been shown to be an internal corruption and fraud case with deliberate tampering of a municipal algorithm and misappropriation of public money. The network traffic, file transfers, and email content are forensically analyzed to provide a complete and coherent narrative connecting the technical evidence to the human actors responsible for it. This investigation thus discloses not only the method of data exfiltration but the criminal intent behind it as well.

4 Appendices